EEI6366 | Lab 01: Big Data Processing with PySpark

Student Handout: Lab Evaluation

1. Form Your Group

- **Maximum 3 members** per group.
- Use the course **Padlet** to find teammates: padlet.com/nmsiy/eei6366-lab01-big-data-processing-with-pyspark-s0250k8sjlyctrys32cu

2. Prepare and Do the Lab

1. Study the **"Getting Started with PySpark"** file.
2. Choose a big dataset (**100,000 rows minimum**) from **Kaggle** or generate one using an **AI tool**.
3. Explore it using **PySpark**: at least **3 operations**.
4. Explore it using **Spark SQL**: at least **3 operations**.

3. Padlet Post

- Each group makes **one post** covering everything you learned from this lab.
- Tip: include the dataset and source, the operations you used, one key finding and one challenge.

4. Peer Review

- Select **3 posts from other teams** and give constructive feedback on each.
- Present this feedback during your viva.

5. Presentation and Viva

- Each team presents, followed by questions. **Every member must present and answer queries.**
- Maximum 10 slides.
- One (01) **member** must select the time slot (LMS choice) on behalf of the team.
- Presentation 09 minutes and 06 minutes for Q/A. (15 Minutes).

Suggested slide outline

1. Title and team members	5. Spark SQL operations (3)
2. Dataset: source, size, why chosen	6. Key findings and challenges
3. Setup and approach	7. Peer review of 3 posts
4. PySpark Operations (3)	8. Conclusion and lessons learned

Key Dates

Padlet Post due: **01st Nov 2026** PPT Due: **02nd Nov 2026** Viva date: **3rd, 4th, 5th, 6th, & 7th Nov 2026**